

HYEONMIN JEON

Financial Investment AI Lab
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Applying for Ph.D., Fall 2027

EDUCATION

Yonsei University

Seoul, Korea

M.S. in Industrial Engineering

Mar 2025 – Feb 2027 (expected)

Advisor: Prof. Kyong Joo Oh, Financial Investment AI Lab. GPA 3.97/4.0.

Relevant coursework: Stochastic Modelling Theory, Probabilistic Models & Bayesian Inference, Time-Series Analysis, Deep Learning Theory & Applications, Advanced Investment Engineering, Financial Engineering Topics, Corporate Finance.

Chung-Ang University

Seoul, Korea

B.S. in Electrical and Electronics Engineering

Mar 2019 – Feb 2025

GPA 3.59/4.0. Thesis: *Light-weighting Tiny YOLOv3: The Impact of Quantization on Real-World Applications* (advisor Prof. Minhyeok Lee).

Relevant coursework: Linear Algebra, Mathematics for Electrical Engineering, Numerical Analysis, Random Variables, Signals & Systems, Machine Learning, Artificial Intelligence, Algorithms.

RESEARCH INTERESTS

Bayesian learning and simulation optimization under input uncertainty; learned and multi-fidelity surrogate models; generative models (diffusion, normalizing flows) as simulation input models; sequential decision-making with statistical guarantees.

PUBLICATIONS AND WORKING PAPERS

Note. *: corresponding author.

JOURNAL MANUSCRIPTS UNDER REVIEW

1. **Hyeon Min Jeon**, Hee Soo Lee*, Kyong Joo Oh*. Conditional Diffusion Transformers for Multi-Asset Scenario Generation and Portfolio Risk Measurement. *Expert Systems with Applications*, under review, 2026.
2. **Hyeon Min Jeon**, Hee Soo Lee*, Kyong Joo Oh*. FinPhasor: Phase-Preserving Measurement of Cross-Sectional Timing in Equity Markets. *Expert Systems with Applications*, under review, 2026.
3. Sang Hoe Kim, **Hyeon Min Jeon**, Dae Hyuk You, Jiyoung Jeon, Seungho Baek, Hee Soo Lee, Kyong Joo Oh*. Deep Convolutional Factor Prediction with Explainable AI for Enhanced Investment Strategy. *In preparation for resubmission*, 2026.
4. Dae Hyuk You, Jiyoung Jeon, **Hyeon Min Jeon**, Sang Hoe Kim, Sang Hyuk Yoo, Tae Yoon Kim, Hee Soo Lee, Kyong Joo Oh*. Benchmark-Aware Enhanced Indexing Guided by the Volume-Price Index. *In preparation for resubmission*, 2026.

WORKING PAPERS

5. **Hyeon Min Jeon**. Learned Surrogate Priors for Bayesian Model Selection: Density, Uncertainty, and the Cost of a Sampler. Lab seminar, January 2026. [\[Link\]](#)
6. **Hyeon Min Jeon**. Bayesian Capacity Sizing for an Exchange Matching Queue under Over-Dispersed Arrivals. 2026. [\[Link\]](#)

CONFERENCE PRESENTATIONS

7. YoungDoo Jeong*, HaeRim Kim, Dae Hyuk You, **Hyeon Min Jeon**, Andre de Castro, Taewoo Lee. A Study for Forecasting Solar Energy Power Generation. *45th International Symposium on Forecasting (ISF 2025)*, Beijing, July 2025. Talk, joint work with SAS Institute.
8. **Hyeon Min Jeon**, Minhyeok Lee. Dealing with the Imbalanced-Data Problem with a Diffusion Model. *KICS Fall Conference*, November 2023.
9. **Hyeon Min Jeon** et al. Personalized Sleep Score Prediction Model Using a Lifelog Dataset. *ETRI Human Understanding AI Paper Challenge*, 2024.

RESEARCH EXPERIENCE

Financial Investment AI Lab, Yonsei University

Graduate Research Assistant

Seoul, Korea
Mar 2025 – Present

- Generative scenario models for portfolio risk: a conditional diffusion transformer that generates joint multi-asset return scenarios, evaluated by the VaR/ES reserves set on its scenarios rather than by sample realism.
- Timing in the equity cross-section: forecasting complex wavelet scalograms so that the phase of price moves, and hence when each stock reprices, is preserved and used for cross-sectional ranking.
- Learned priors for Bayesian factor selection: replacing the mixture-of- g prior with a normalizing flow that keeps an explicit density, checked against the exact posterior over models (TV distance 0.016) and tuned by Bayesian optimization; compared with a density-free GAN prior.
- Ranking and selection under input uncertainty: implemented and verified KN, feasibility determination, recycled observations and CRN, and studied how a fitted input model affects the probability-of-correct-selection guarantee.

TIPS R&D Project with Urban Complex

Researcher, artist-style generative engine (lab project)

Seoul, Korea
Mar 2025 – Aug 2026

- Built chain-of-thought style captioning with Qwen and trained a QLoRA style classifier on Qwen2-VL-7B (91.4% seen / 76.0% unseen; 96.0% fused with a DINOv2 prior).
- Trained SDXL-based generation with domain checkpoints routed by the classifier, and built the style memory module.

GAIL (AI Lab), Chung-Ang University

Undergraduate Researcher

Seoul, Korea
Dec 2022 – Dec 2024

- Diffusion-model oversampling for imbalanced fraud data; latent-diffusion portfolio study; lifelog sleep-score prediction; 3D point-cloud classification.

Carbon-Neutral ESG Research Group, Chung-Ang University

Team Lead, “Green-Navi” (Green AI track)

Seoul, Korea
Apr 2023 – Dec 2023

- Reinforcement learning for fuel-optimal urban routing in AIMSUN; presented at the research fair, November 2023 (MSIT/NRF Field-Linked Future-Leading Talent programme).

SELECTED PROJECTS

Retina Exam Room Scheduling, with Nune Eye Hospital

Discrete-event simulation and reinforcement learning

Feb 2026 – Present

- Recalibrated a discrete-event simulator to hospital records; compared rule, behaviour-cloned and PPO policies on a 44-day replay (time in room 12.2 $\rightarrow$ 8.2 min); read-only EMR integration and live comparison.

Simulation-Methodology Reimplementations

rs-lab, Input-Uncertainty Select, TailFlow, Eco-Route RL, roughvol-lab

Sep 2026

- Ranking & selection with verified guarantees; input-uncertainty effects on selection; conditional diffusion as a risk engine; CO₂-optimal routing on SUMO with P(correct selection) 0.956; rough

volatility.

Lucubra, research-agent platform

LLM agents for paper verification and referee reports

Sep 2026 – Present

Light-weighting Tiny YOLOv3

Graduation thesis, team of three

2024

- Int8 quantization with layer fusion; total footprint 1.6 GB $\rightarrow$ 100 MB, inference 22–64% faster across five devices, mAP -2.78 pt on PASCAL VOC.

TEACHING EXPERIENCE

Yonsei University

Seoul, Korea

Teaching Assistant, Graduate Program in AI Investment Engineering

Sep 2025 – Present

Yonsei University

Seoul, Korea

Teaching Assistant, Financial Engineering (undergraduate)

Spring 2026

AWARDS AND HONORS

- Special Prize (data use), Chungcheong ICT Innovation Public-Data Competition 2024
- DB Financial Economics Paper Competition, national-pension product proposal 2024

SKILLS

PROGRAMMING	Python (NumPy, SciPy, pandas, PyTorch, CUDA), C++, MATLAB, TypeScript, SQL, $\text{\LaTeX}$
METHODS	Discrete-event and Monte Carlo simulation, ranking & selection, variance reduction, diffusion and flow models, Bayesian inference, reinforcement learning, time-series analysis
TOOLS	Eclipse SUMO, AIMSUN, SAS, Hugging Face, Linux multi-GPU servers
LANGUAGES	Korean (native), English (working proficiency)